Text Mining for AI 2026

Lecture 1: Introduction to Text Mining

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Outline

- Introduction to Text Mining
- Course structure and assignments
- Examples of Text Mining applications

There is no such thing as simple text!

There is no such thing as simple text

twitterer

The new US president has said he believes that vaccines are harmful and has repeatedly and erroneously suggested that they cause autism

There is no such thing as simple text

twitterer

The new US president has said he believes that vaccines are harmful and has repeatedly and erroneously suggested that they cause autism

The new US president

A noun phrase, refers to a person. Who is this person? Contextual info needed to figure out that this expression refers to a particular person.

said he believes

Words point to a source (sb said smth) and indicate that this person believes in some kind of proposition.

vaccines are harmful

What does this person believe: a statement (claim).

repeatedly and erroneously suggested

A clear judgment of the twitterer with respect to the source of the claim (‘vaccines are harmful’), which is US president, which is referring to one particular person.

they cause autism

Another claim. What does ‘they’ represent here? We (humans) will connect ‘they’ to vaccines.

Short text = complex massage with a lot of relations and information.
-> NLP technology to extract this information.

There is no such thing as simple text

twitterer

The new US president has said he believes that vaccines are harmful and has repeatedly and erroneously suggested that they cause autism

The new US president

Entity phrase detection

Entity Linking

said he believes

provenance attribution (links claim to source)

coreference (he refers to president)

vaccines are harmful

semantic parsing (harmful = property of vaccines)

repeatedly and erroneously suggested

provenance attribution

judgement

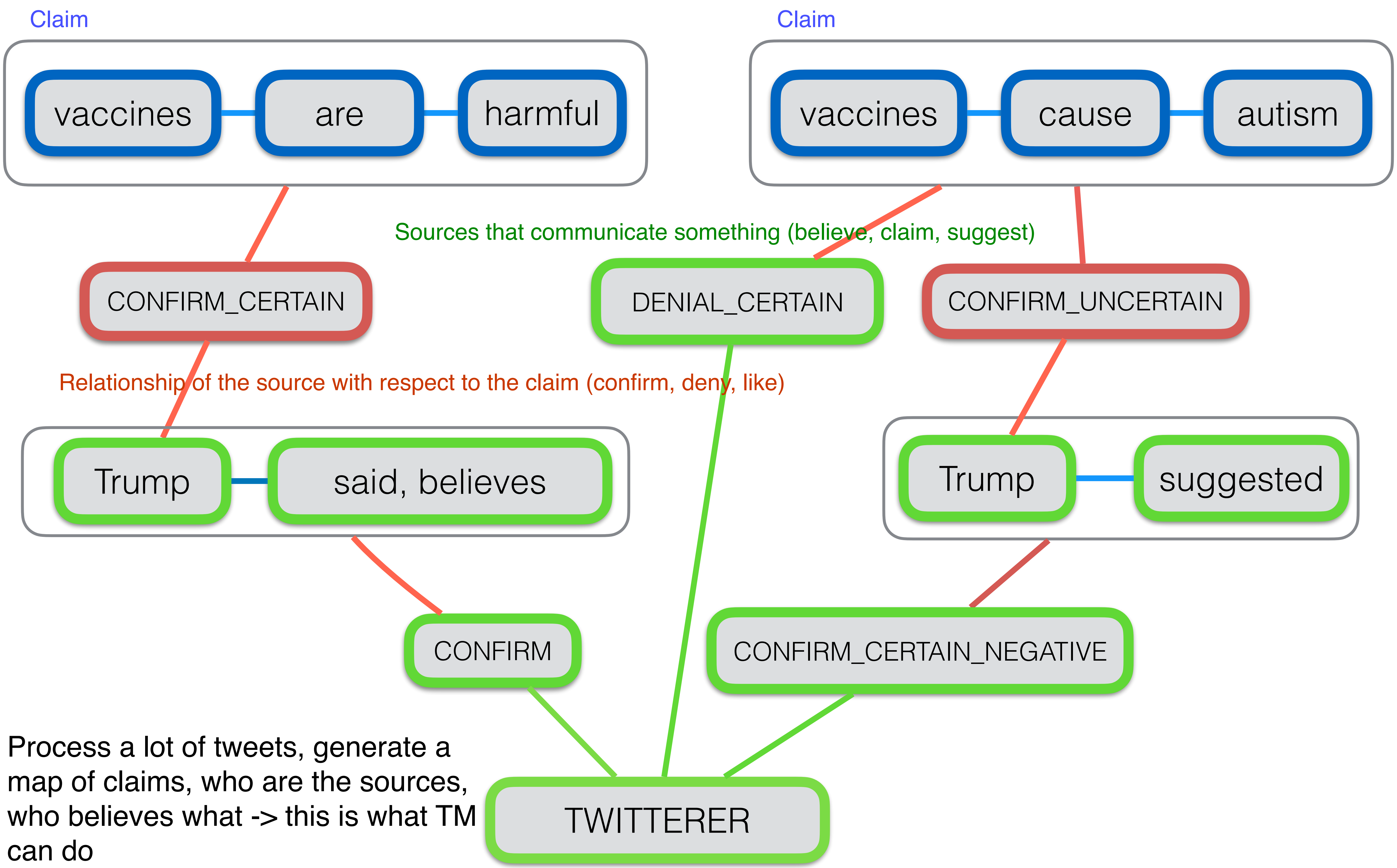
they cause autism

semantic parsing

coreference

All these pieces of analysis generate the information that is implied by this short text, but in fact this represents a complex graph of information and knowledge

Perspective Graph



Process a lot of tweets, generate a map of claims, who are the sources, who believes what -> this is what TM can do

Terminology

- **Computational linguistics**: algorithms that model language data and define notions, e.g. *similarity, information value, sequence probabilities, language models*
- **Natural Language processing** (NLP): engineering to address aspects of natural language, e.g. *tokenisation, lemmatisation, compound splitting, syntactic parsing, entity detection, sentiment analysis*
- **NLP Toolkits**: software packages and resources that provide and/or combine collections of NLP modules
- **Language applications**: machine translation, summarisation, chat bots, text mining
- **Text mining**: from unstructured text to structured data (information or knowledge) (our focus: understand the technology, limitations, build applications)

NLP Toolkits (Python)

- NLTK: http://www.nltk.org (well documented, data, multiple languages)
- spaCy: https://spacy.io (faster, industrial use)

- AllenNLP: https://allennlp.org (focuses on deep learning)
- Huggingface: https://huggingface.co (Google, provides tools, data, models)
- These toolkits solutions for: tokenisation, part-of-speech tagging, syntactic parsing, named entity detection, language models, and more for many languages
- **Utility packages** (software packages underneath these toolkits)**:**
    - Scikit-learn, Gensim, Pandas, etc…..

Course Goals

- Introduce the field of Natural Language Processing (NLP)
  - theory & technologies
  - possibilities & limitations in the real world
- Provide you with the practical knowledge to perform basic NLP tasks on text using off-the-shelf tools
- Resulting in the ability to carry out an experiment involving basic NLP tasks, evaluate the results and present a balanced report (poster) about the experiment

Course Setup

- Every week one lecture & one practical session (except for this week (one lecture only) and next week in which we have two lectures)

- Theoretical aspects of text mining will be covered during the lectures

- Lab sessions (Jupyter notebooks) and assignments to gain practical text mining skills and apply the theoretical concepts

- Text mining project in which you apply the skills to solve a problem or a task

Course set up (see Canvas)

- **Lectures** (typically Tuesdays, see below), Lab sessions (typically Fridays, see below)
  - **Week 1**: Introduction to text mining (31 March)
    - ***Groups formed (Monday 6th)*** (join a group on Canvas, max 4 students)
  - **Week 2**: Linguistics & Natural Language Processing and Machine Learning I (7- April & 10-April) & **Lab-1** (10 - April)
    - ***Lab1-Assignment-toolkits (13 - April)*** (basing processing with NLTK and SpaCy)
  - **Week 3**: Natural Language Processing & Machine Learning II (14 - April) & **Lab-2** (17 - April)
    - ***No assignment***
  - **Week 4**: Sentiment (21 - April), **Lab**-**3** (24 - April)
    - ***Lab3-Assignment-sentiment (4 - May)***,
  - **Week 5**: Entity Recognition (8 - May), **Lab**-**4** (8 - May)
    - ***Lab4-Assignment-entities (11 - May)***
  - **Week 6**: Text categorisation & Topic Modelling (12 - May), **Lab**-**6** (13 - May)
    - ***Lab6-Assignment-topic-modelling (18 - May)***
    - **Text for the project** (group project)
  - **Week 7:** Final questions (19 - May)
    - ***Multiple Choice exam: May 27***
    - ***Project report: June 1***
- **Submissions**: All assignments are submitted through CANVAS as a ZIP file that includes both a PDF and the notebook. If you fail to convert a notebook to PDF you can also submit just the notebook after contacting your TA. **Always include the group number and student names. This is compulsory** (see Assignment 1 for the format: Group<GROUP_NUMBER>_lab<LAB_NUMBER>, e.g., Group1_lab1). Each group is assigned 1 TA.

Grading

- Group:
  - 4 lab assignments (pass/no pass) (evaluation covers various aspects, e.g, critical view on this technology, comparing outputs of different systems)
  - Project report (group, 40% of your grade)
- Individual:
  - Multiple choice exam (individual, 60% of your grade)
  - Lecture slides
  - Literature as background (to have a more in-depth understanding of the concepts)
  - Weekly self tests on Canvas, which counts as your practice exam (not compulsory)
- For both parts, your grade should be a 5 or higher in order to pass, average should be 5.5 or higher.

About the groups

- Join a Canvas group by the end of this week:
  - Lab sessions start April 10th:
    - 13:30 - 15:15
  - 4 students max, we will merge smaller groups into groups of 4 by **Monday 6th!**
- Each group will be assigned to a TA
- Raise your hand and ask questions when a TA comes by to get feedback
- Notebook assignments are submitted as a group ***but*** do the notebooks individually to acquire the skills and do your part in the group project

Weekly lab assignments

- 4 assignments in total:
  - Lab1-Assignment-toolkits (13 - April)
  - Lab3-Assignment-sentiment (4 - May)
  - Lab4-Assignment-entities (11 - May)
  - Lab6-Assignment-topic (18 - May)

  No lab 2 and no lab 5 this year.
- Completing the weekly lab assignments is a requirement to pass the course
- Resubmission rules for lab assignments: you are allowed to resubmit only two weekly lab assignments (with a deadline of one week); if a group member did not contribute to a lab assignment, they must resubmit it individually within one week.

Labs

- Download from: https://github.com/cltl/ba-text-mining
- Tools:
  - Anaconda platform: https://anaconda.org
  - NLTK: https://www.nltk.org
  - SpaCy: https://spacy.io
  - Scikit-learn: https://scikit-learn.org/stable
  - Exercises & assignments as Jupyter notebooks (https://jupyter.org)

Group project

- **Text mining project:**
  - You get a test set in week 6
  - You need to solve three text mining tasks: **sentiment**, **entities**, **topics**.
    - Sentiment analysis and topic analysis should be done at the **sentence level**
    - For one of the tasks, select a range of different **systems/models (2 minimum)**

Group project

- **A text mining project:**
  - Motivate your experiments based on models' performance in relation to their design
  - If your focus is on supervised classification, find a dataset that serves the purpose (motivate that this dataset is appropriate for testing your hypotheses); examples: see “Datasets for final projects” on Canvas module Final Assignment
  - Motivate and define a number of different models or approaches:
    - Vary training data, preprocessing steps, feature representation, model parameters
  - Thorough error analysis of the obtained results!
  - Submit a poster with your findings

From text to knowledge

Text Mining applications

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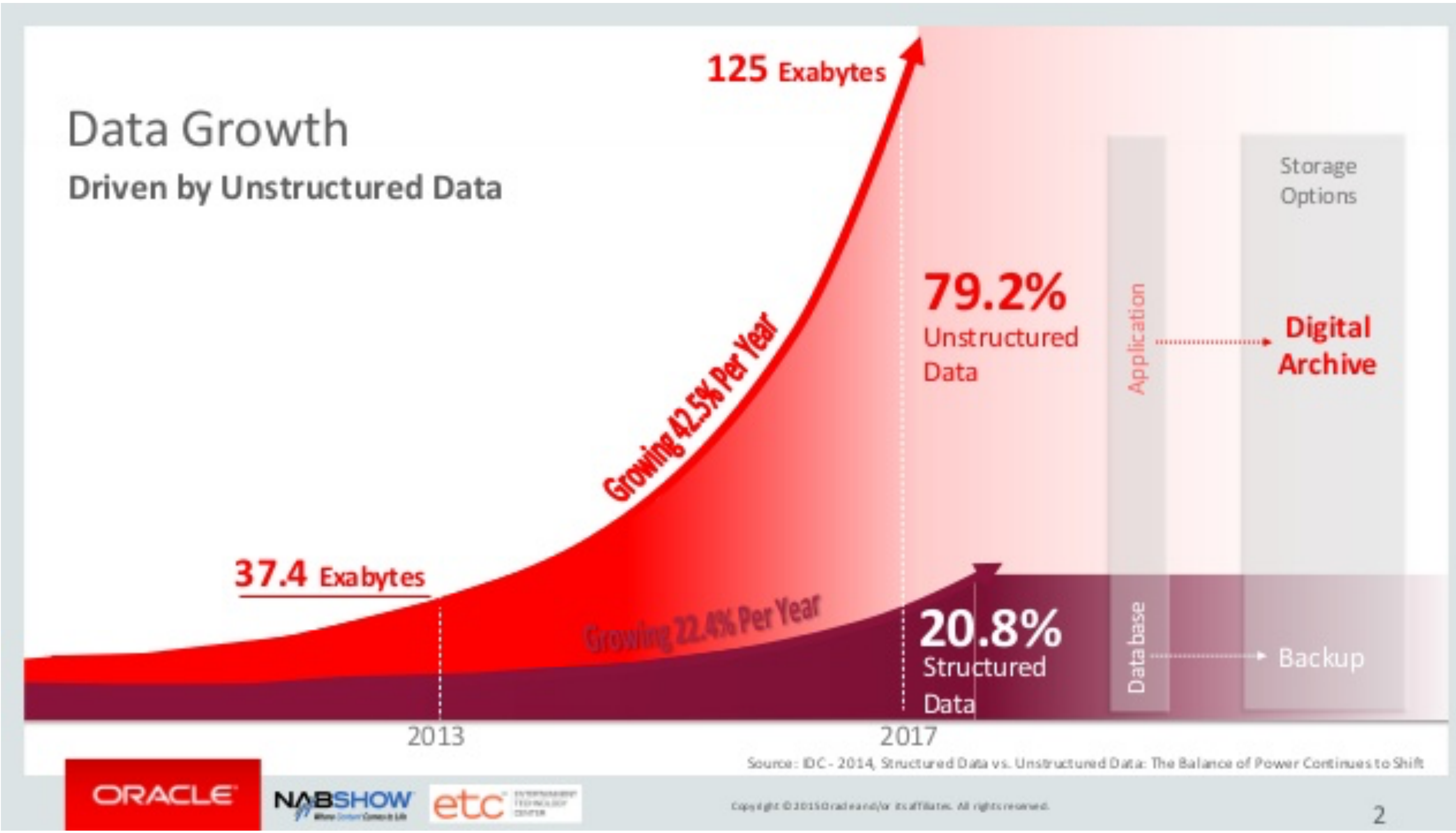
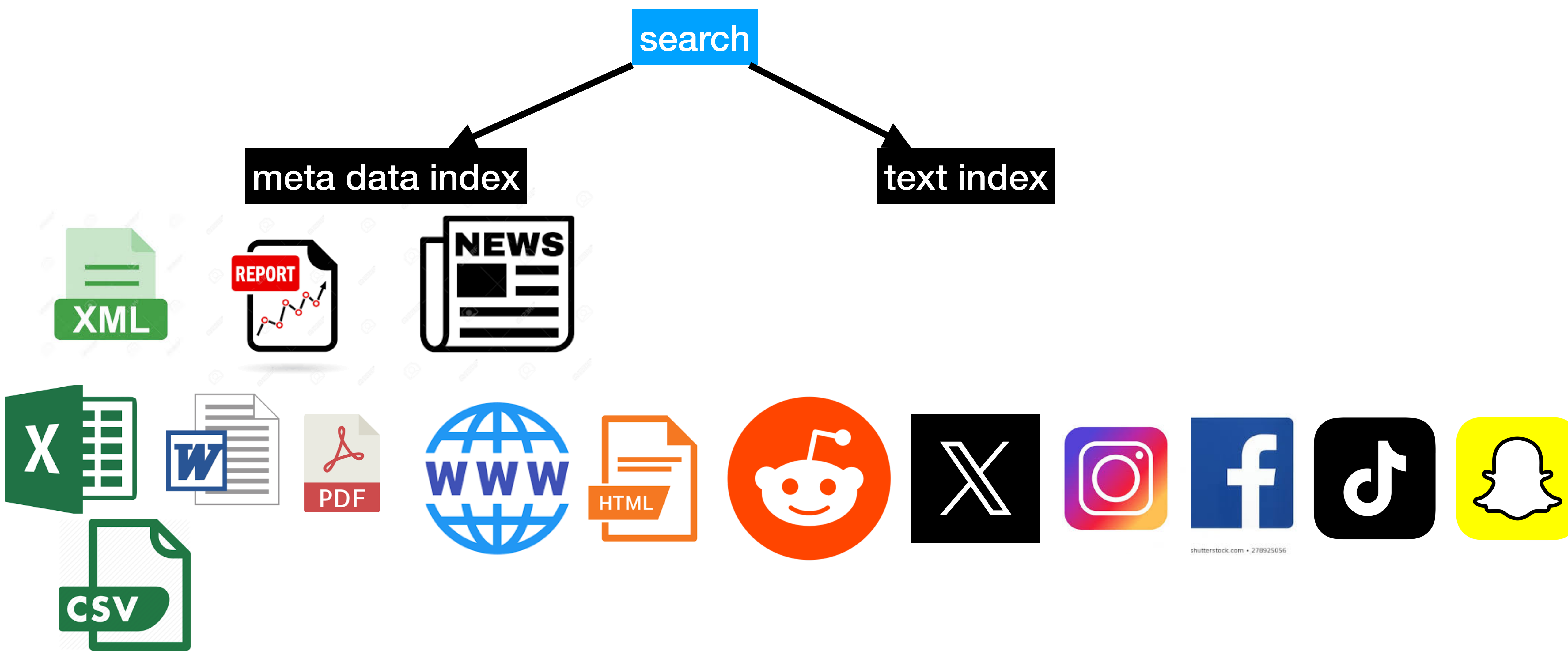
2016
Oracle: we only started to exploit unstructured data



https://www.youtube.com/watch?v=qGLj-I1KxOI&feature=youtu.be&t=1s

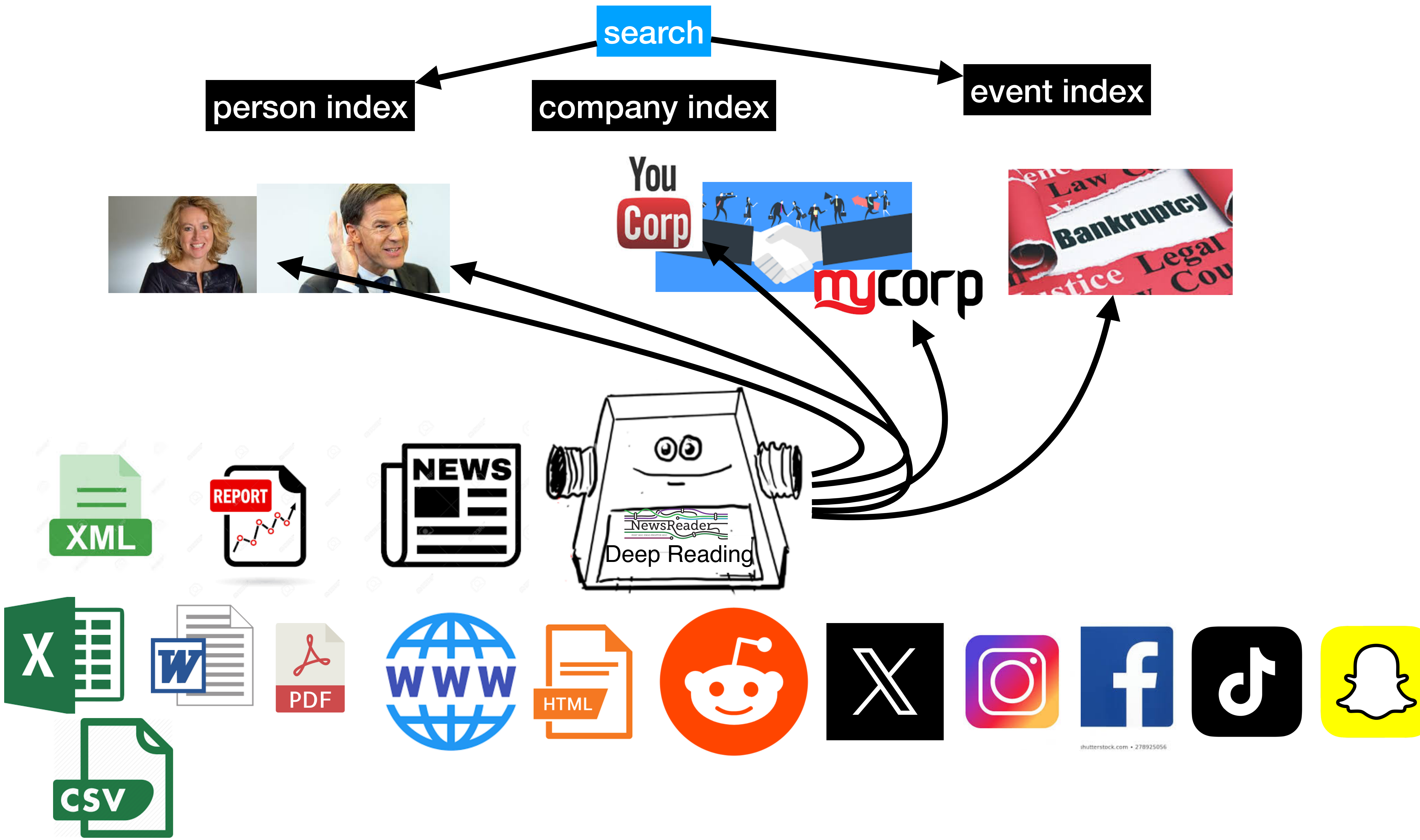
Information pyramid

Information cake

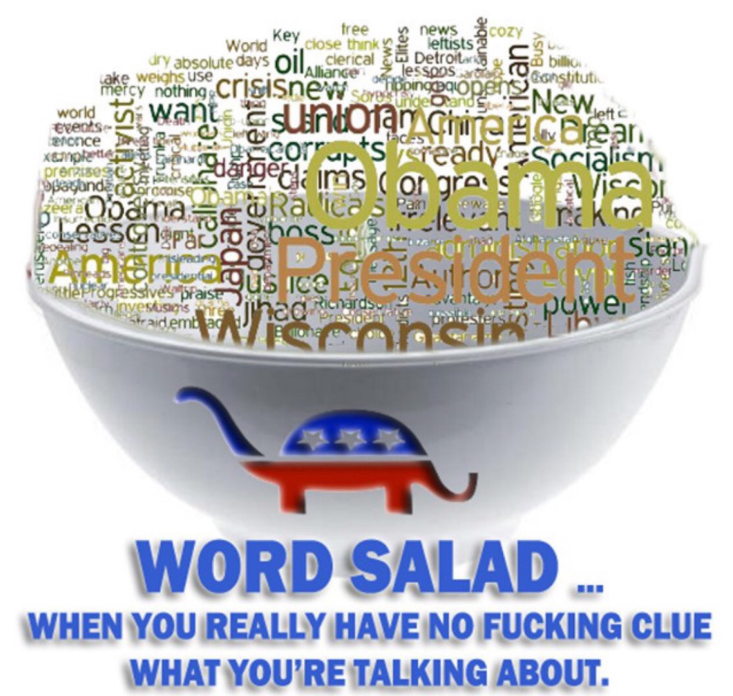


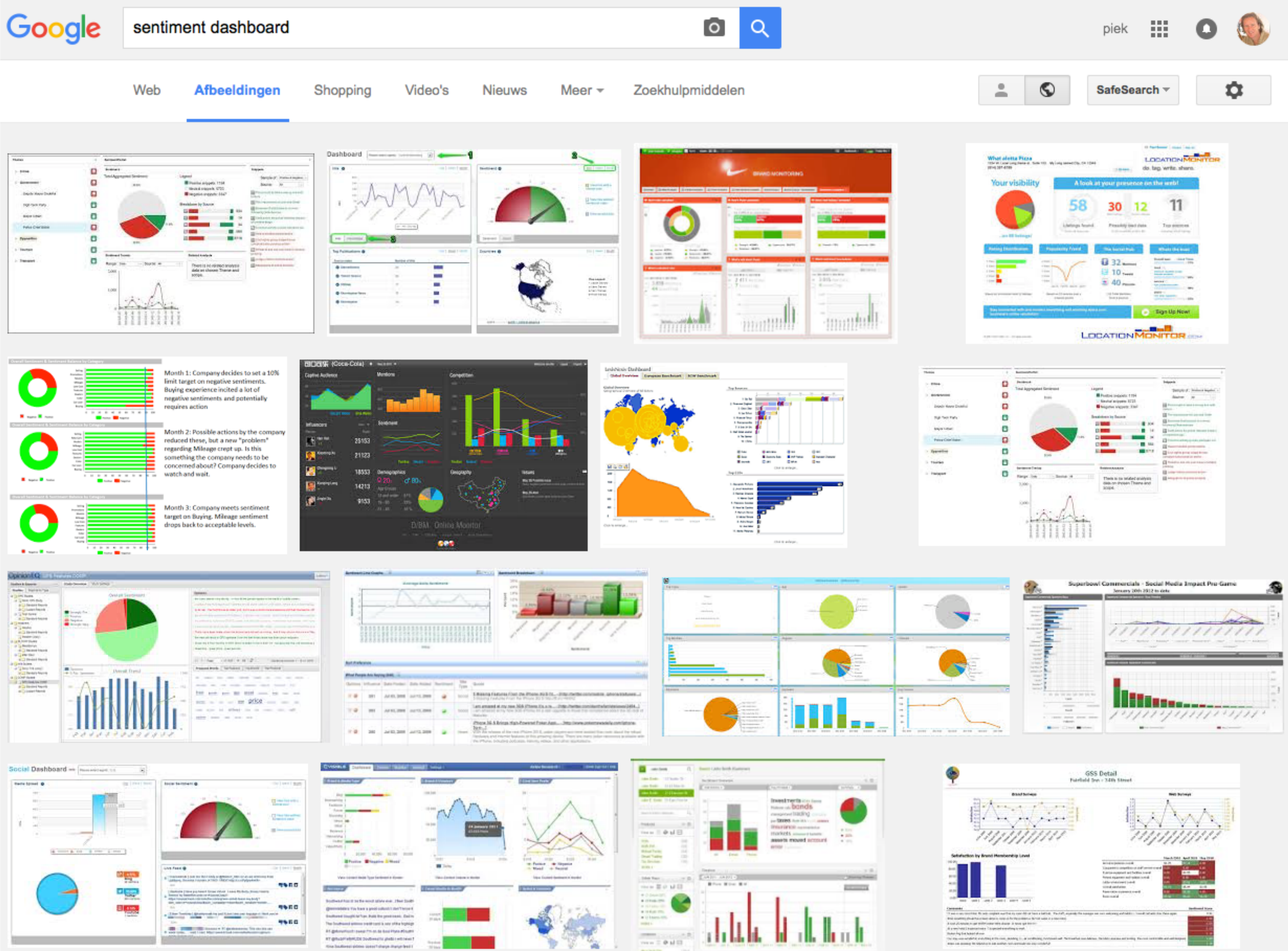
Information cake

Information cake

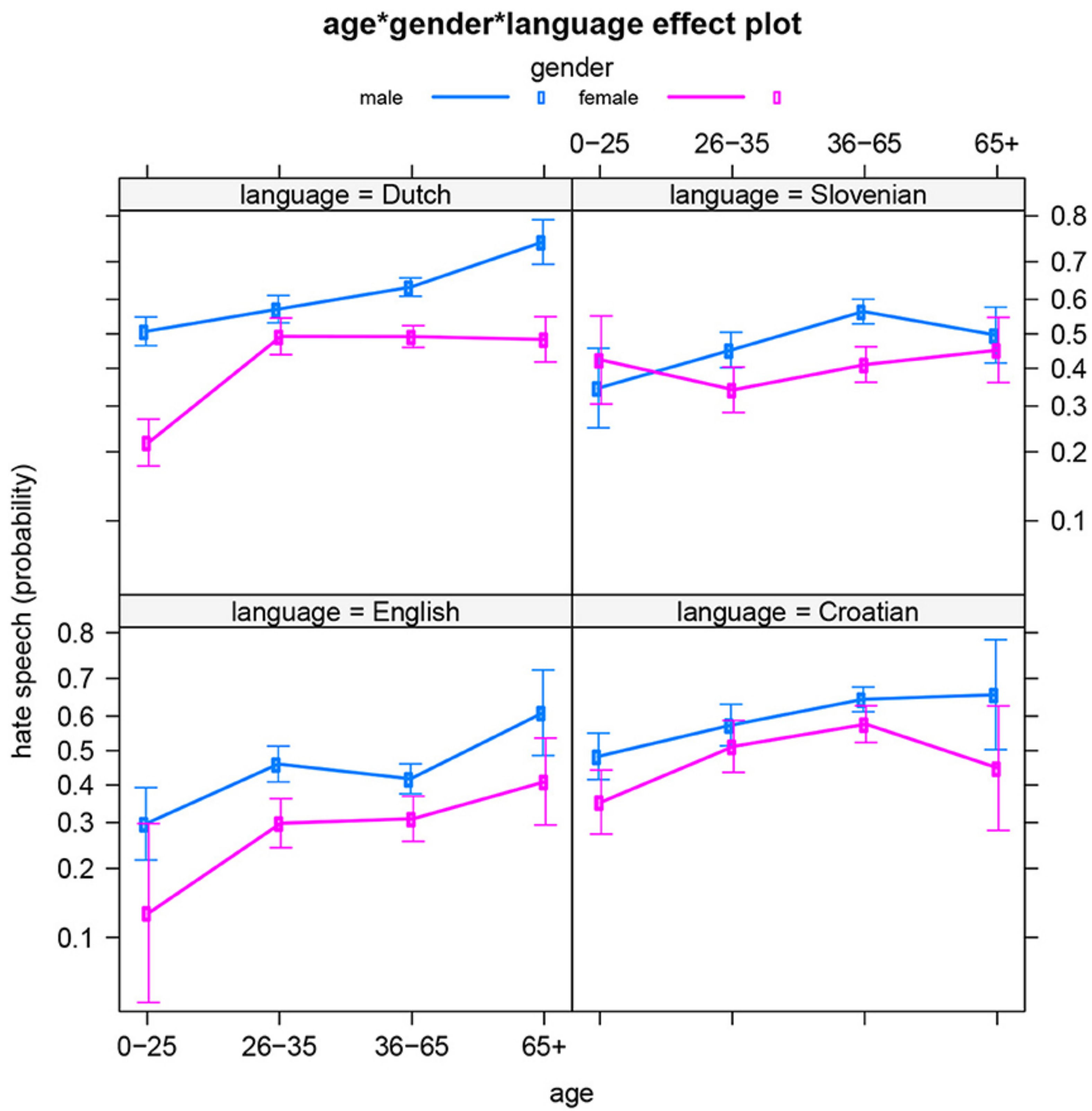


Text mining
Failure and Success

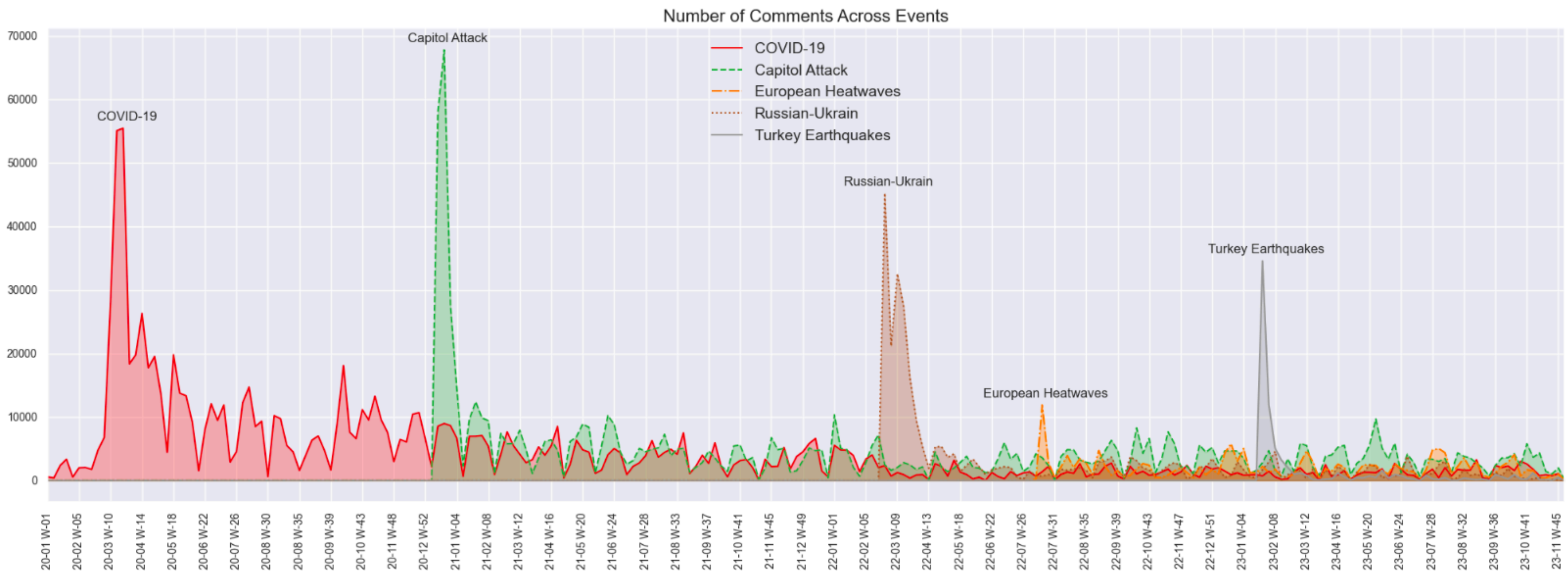



Google
sentiment dashboard
piek
Web
Afbeeldingen
Shopping
Video's
Nieuws
Meer
Zoekhulpmiddelen
SafeSearch
Dashboard
Your visibility
A look at your presence on the web!
58
30
12
11
Sign Up Now!
Month 1: Company decides to set a 10% limit target on negative sentiments. Buying experience incited a lot of negative sentiments and potentially requires action
Month 2: Possible actions by the company reduced these, but a new "problem" regarding Mileage crept up. Is this something the company needs to be concerned about? Company decides to watch and wait.
Month 3: Company meets sentiment target on Buying. Mileage sentiment drops back to acceptable levels.
Superbowl Commercials - Social Media Impact Pre-Game
Social Dashboard
GSS Detail

Who are the haters? A corpus-based demographic analysis of authors of hate speech



Grounding Toxicity in Real-World Events across Languages



(a) Temporal distribution of comment across major world events. The distribution shows aggregate comments from all languages

(b) Toxicity of comments for each event. We computed proportional toxicity by dividing the number of toxic comments by the total number of comments in a particular period.

National Science Agenda

- **11K** questions with titles, descriptions and keywords
- Very diverse:
  - different topics and areas
  - layman and professional language
- Term extraction & Hierarchical clustering
- http://i.amcat.nl/renwa/index.html

NWO Wetenschapsagenda VU Topic Browser

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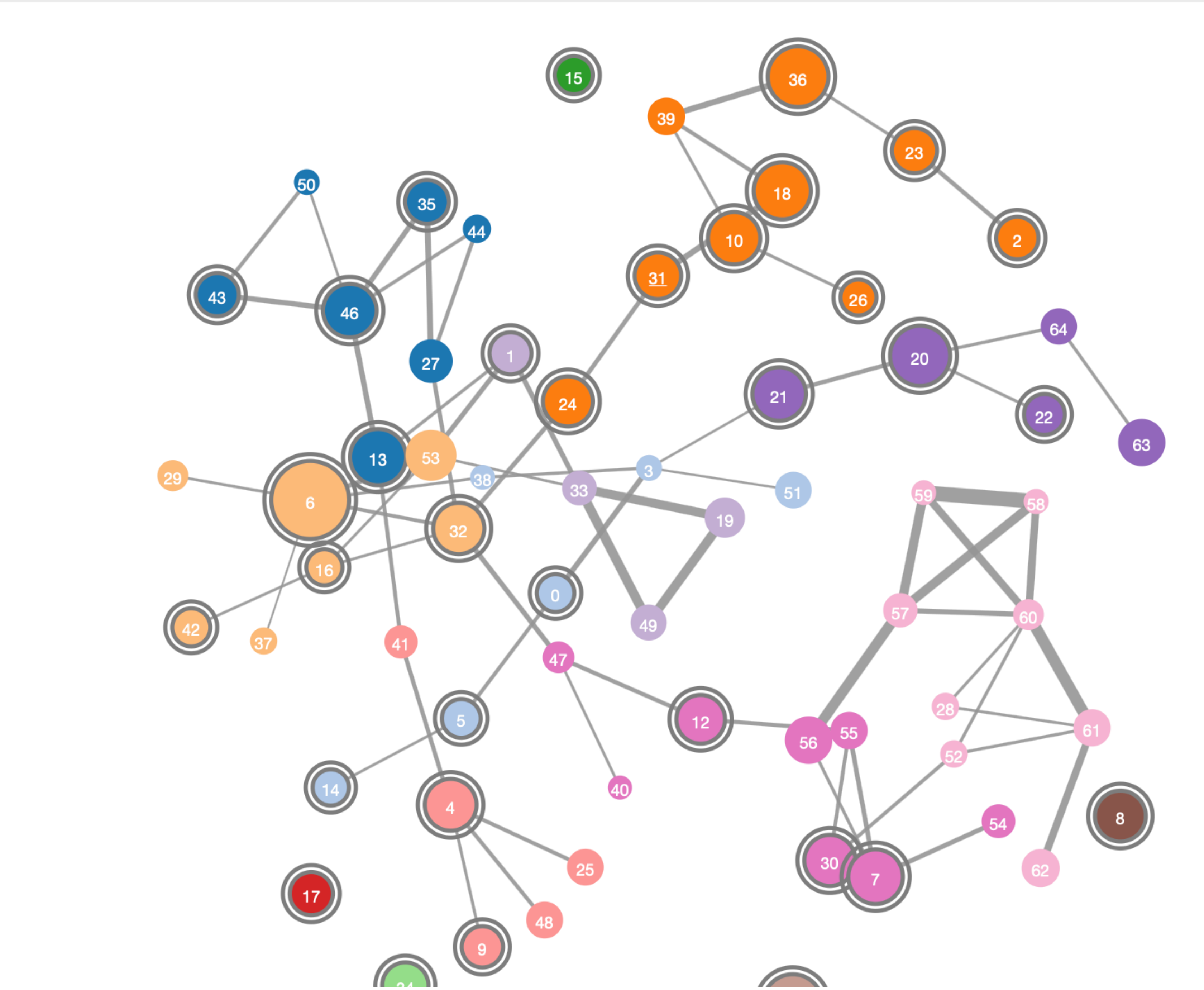
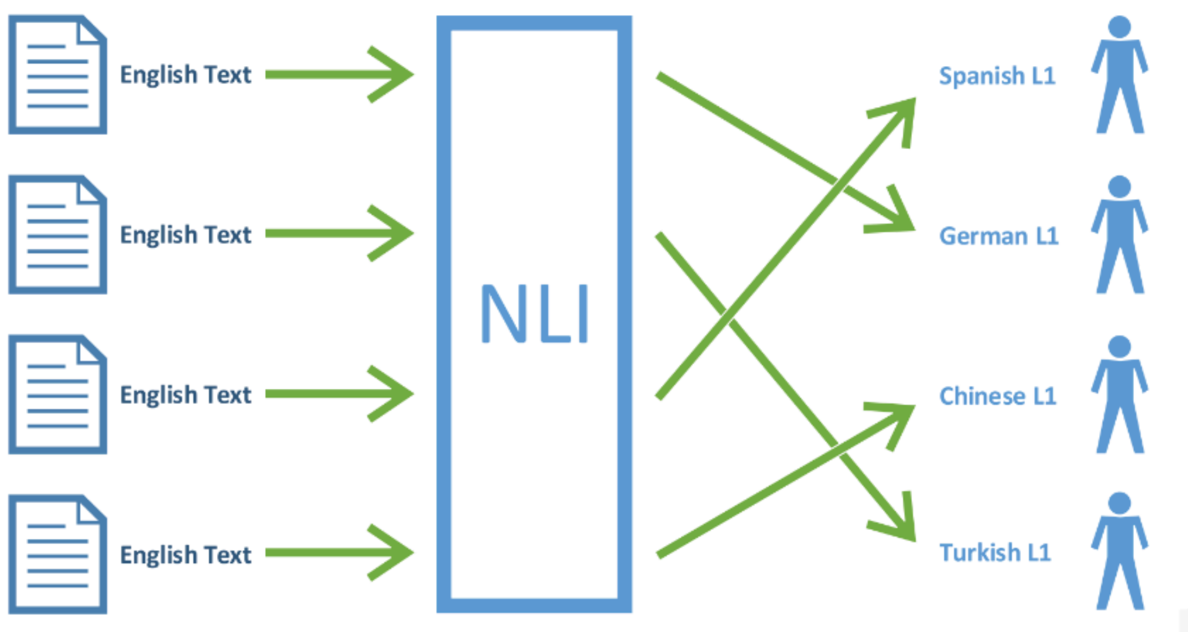
Emiel van Miltenburg, Kasper Welbers, Hennie van der Vliet, Wouter van Atteveldt, Piek Vossen

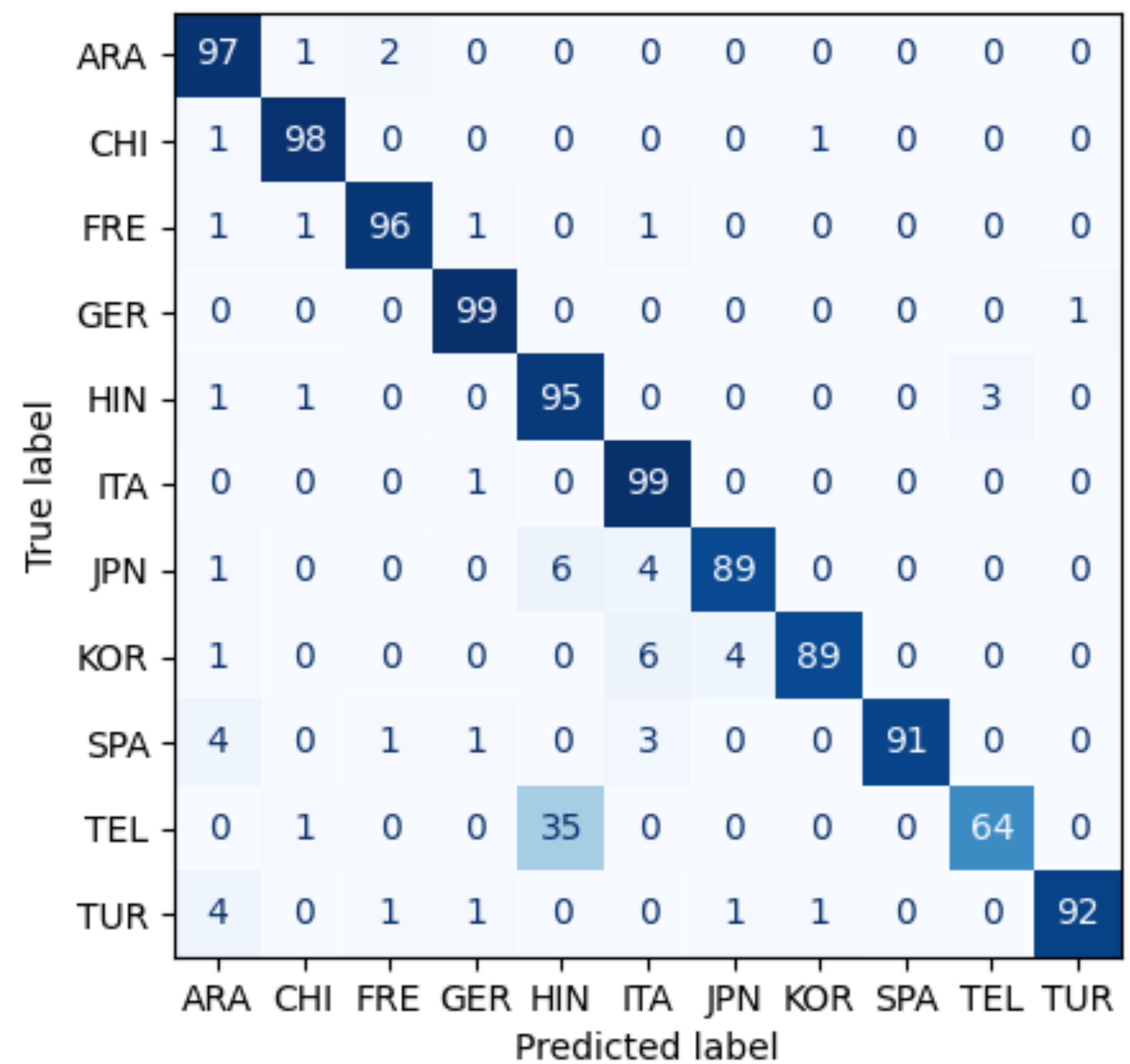


Selected cluster

Selected cluster:31
#Questions: 133
Top words: fossiele_brandstof, fossiele_brandstoffen, brandstoffen, energie, opwarming, opwarming_van_de_aarde, aarde, duurzame_energie, energiebronnen, brandstof

Native Language Identification





Vaccinpraat: Monitoring Vaccine Skepticism in Dutch Twitter and Facebook Comments

(a) Entities (red=org, blue=loc, grey=per) (b) Hashtags

Figure 1: Discussion topics in vaccine-skeptic messages.

Vaccinpraat: Monitoring Vaccine Skepticism in Dutch Twitter and Facebook Comments

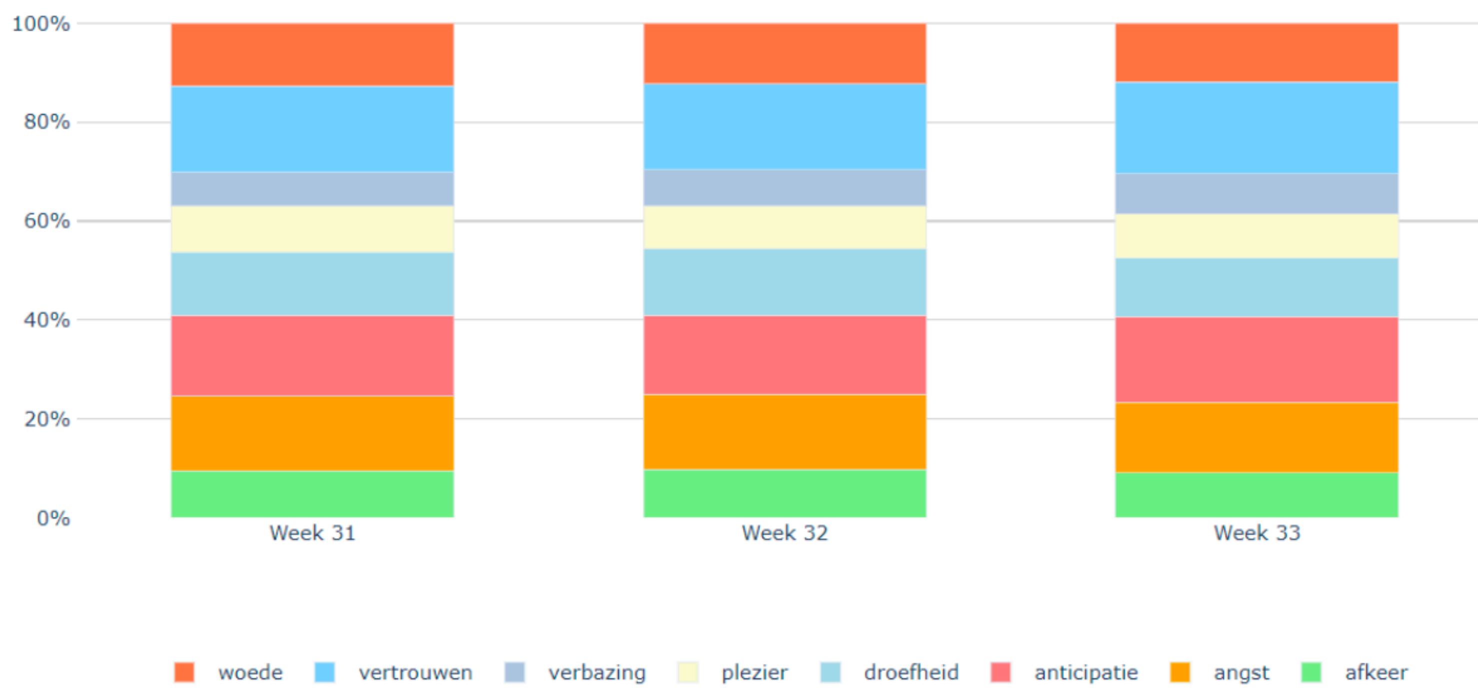


Figure 2: Relative frequencies of the most dominant emotion per message in weeks 31-33 of 2021.

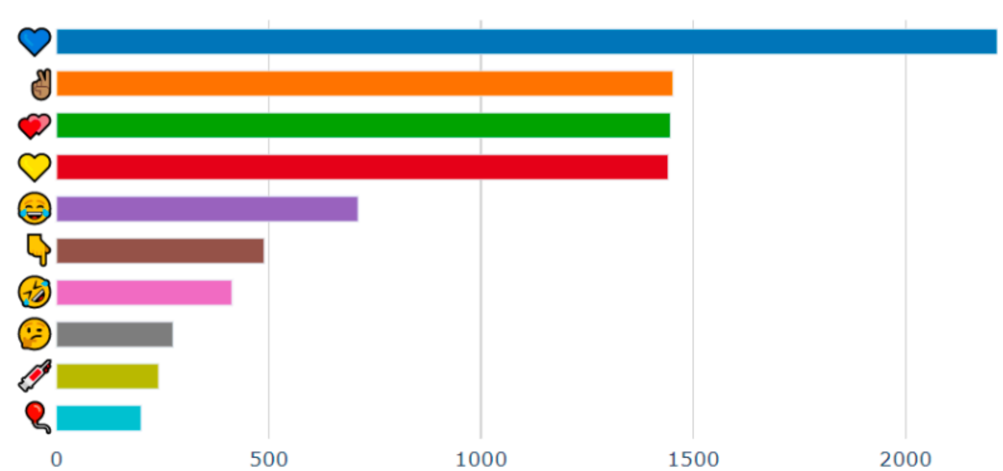
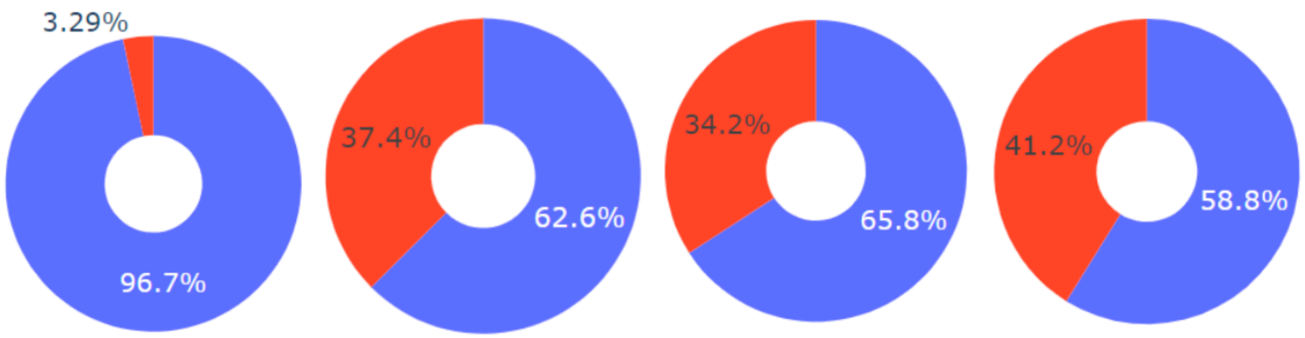


Figure 3: Absolute frequencies of the most frequently occurring emoji in vaccine-skeptic messages.

Vaccinpraat: Monitoring Vaccine Skepticism in Dutch Twitter and Facebook Comments

Figure 4: Heatmap of Twitter users with anti-vaccination opinions.



(a) Age (blue=adult, red=adolescent

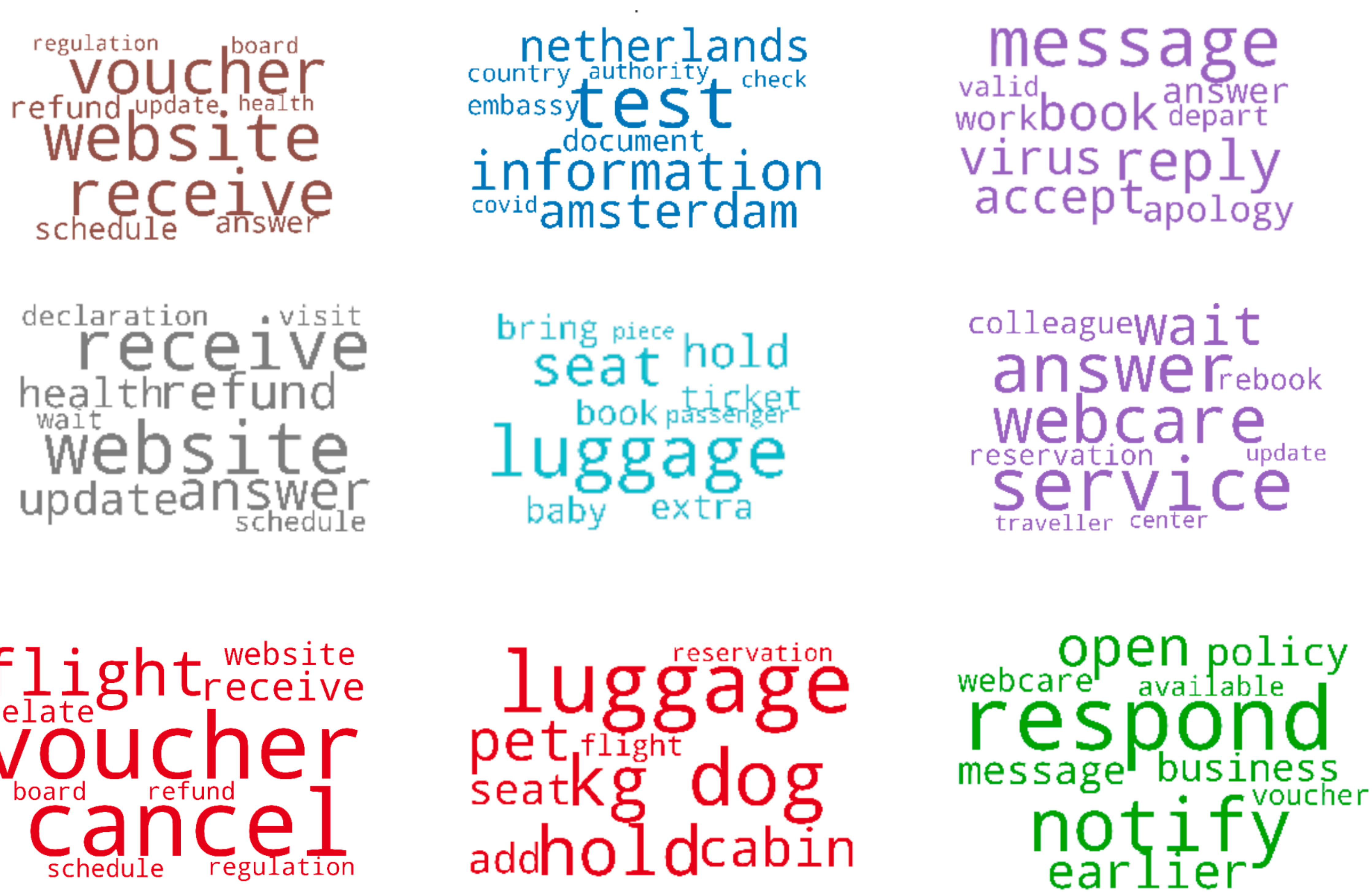
(b) Gender (blue=male, red=female)

(c) Personality (blue=extrovert, red=introvert)

(d) Education level (blue=low, red=high)

Figure 5: Authorship attributes of vaccine-skeptic Twitter users.

Automatic Retrieval of Topics Using Topic Modelling Techniques from Customer Conversations in the Airline Domain



Text Mining failures

- **Identities mismatches**
- **Hallucinations** (a tendency to invent facts in moments of uncertainty):
  - Yuri Gagarin was the first person to land on the Moon (the correct answer is Neil Armstrong)
- **Biases**:
  - “man” is closer to “doctor” and “woman” is closer to “nurse”, or “computer programmer” is to “man” as “homemaker” is to “woman”
  - hate speech detection: tweets by African Americans are up to two times more likely to be labelled as offensive compared to others

Master’s programs
Faculty of Humanities

- **Language and AI** (1 year)
- **Research Master’s in Linguistics** (Human Language Technology track) (2 years)